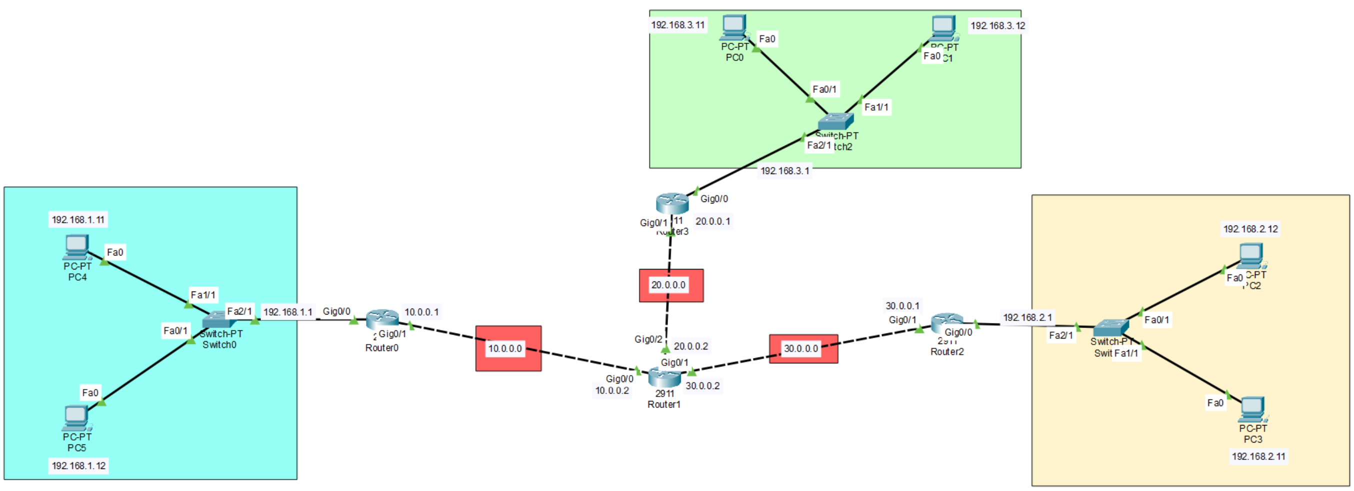
DAY_5

❖ **Configuring OSPFPractical:Configure OSPF on a network with multiple areas.Task: Simulate a scenario where a router fails and observe OSPF reconvergence.**

1. Using Cisco Packet Tracer, create a network topology with three routers in two OSPF areas (Area 0 and Area 1), and assign appropriate IP addresses to all interfaces.

ANS :

➢ To configure **OSPF (Open Shortest Path First)** on a network containing **Area 0 and Area 1**, verify OSPF neighbor relationships, test connectivity between PCs, and observe OSPF behavior during a link failure and recovery.



Device	Interface	IP Address	Network	OSPF Area
R1	G0/0	10.0.0.1/30	10.0.0.0/30	Area 0
R2	G0/0	10.0.0.2/30	10.0.0.0/30	Area 0
R1	G0/1	10.0.1.1/30	10.0.1.0/30	Area 1
R3	G0/0	10.0.1.2/30	10.0.1.0/30	Area 1
R2	G0/1	192.168.10.1/24	192.168.10.0/24	Area 0
R3	G0/1	192.168.20.1/24	192.168.20.0/24	Area 1
PC1	NIC	192.168.10.10/24	192.168.10.0/24	Area 0
PC2	NIC	192.168.20.10/24	192.168.20.0/24	Area 1

2. **Configure OSPF on each router in your topology so that Area 0 acts as the backbone and Area 1 connects through one of the routers; verify OSPF neighbor relationships using the 'show ip ospf neighbor' command.**

ANS:

- **Router1**

```
enable
configure terminal

hostname R1

interface gigabitEthernet 0/0
ip address 10.0.0.1 255.255.255.252
no shutdown
exit

interface gigabitEthernet 0/1
ip address 10.0.1.1 255.255.255.252
no shutdown
exit

router ospf 1
router-id 1.1.1.1

network 10.0.0.0 0.0.0.3 area 0
network 10.0.1.0 0.0.0.3 area 1

end
write memory
```

- **Router2**

```
enable
configure terminal

hostname R2

interface gigabitEthernet 0/0
ip address 10.0.0.2 255.255.255.252
no shutdown
exit
```

```
interface gigabitEthernet 0/1
ip address 192.168.10.1 255.255.255.0
no shutdown
exit

router ospf 1
router-id 2.2.2.2

network 10.0.0.0 0.0.0.3 area 0
network 192.168.10.0 0.0.0.255 area 0

end
write memory
```

- **Router3**

```
enable
configure terminal

hostname R3

interface gigabitEthernet 0/0
ip address 10.0.1.2 255.255.255.252
no shutdown
exit

interface gigabitEthernet 0/1
ip address 192.168.20.1 255.255.255.0
no shutdown
exit

router ospf 1
router-id 3.3.3.3

network 10.0.1.0 0.0.0.3 area 1
network 192.168.20.0 0.0.0.255 area 1

end
    write memory
```

3. **Add two PCs to different areas in your OSPF network, assign them IP addresses, and test connectivity by pinging from one PC to the other.**

ANS:

- **PC1 :**

  IP Address: 192.168.10.10
  Subnet Mask: 255.255.255.0
  Default Gateway: 192.168.10.1

- **PC2 :**

  IP Address: 192.168.20.10
  Subnet Mask: 255.255.255.0
  Default Gateway: 192.168.20.1

- **Verify the routing table**

  - On R2:

    show ip route
    You should see an OSPF route toward:
    192.168.20.0/24

  - On R3:

    show ip route
    You should see an OSPF route toward:
    192.168.10.0/24

4. **Simulate a router failure by shutting down the interface connecting Area 1 to Area 0, then use 'show ip route' and 'show ip ospf' commands to observe how OSPF reconverges and restores connectivity after re-enabling the interface. Hint: Track the time it takes for OSPF to update the routing tables after the failure and recovery.**

ANS :

- **Shut down the R1 interface**

```
enable
configure terminal

interface gigabitEthernet 0/1
shutdown

    end
```

➢ Check OSPF Neighbor

- On R1:
  show ip ospf neighbor
  The R3 neighbor should disappear after OSPF detects the failure.

- On R3:
  show ip ospf neighbor
  The adjacency with R1 should also disappear.

➢ Check the Routing Table

- On R1:

  show ip route
  The route toward:
  192.168.20.0/24
  will no longer be available through the failed link.

- On R2:

  show ip route
  The route toward the Area 1 network will also be removed because there is no longer a path to Area 1.
  You can inspect OSPF information with:
  show ip ospf
  and:
  show ip ospf interface